

NEPA 3D Knowledge Graph Review System

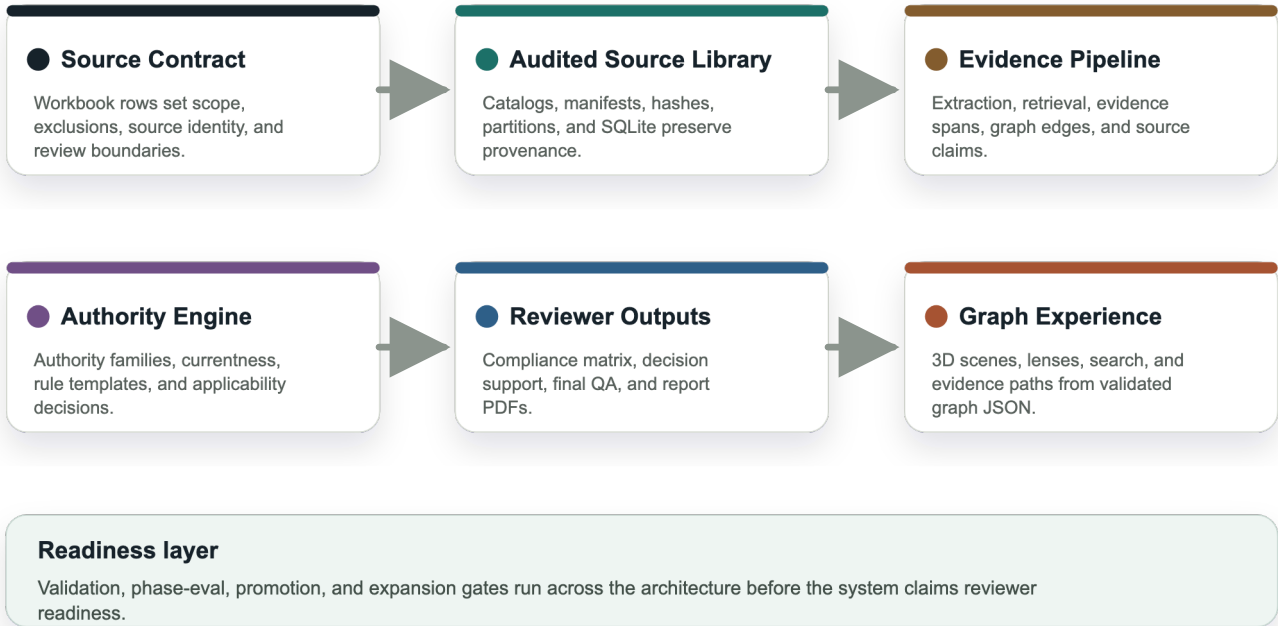
The system turns Region 1 NEPA source material into auditable review intelligence. It preserves source provenance, builds an authority graph, traces evidence to findings, and gates readiness before outputs reach reviewers. The result is a transparent operating layer for compliance review, decision support, and graph exploration.

Current Region 1 implementation snapshot.



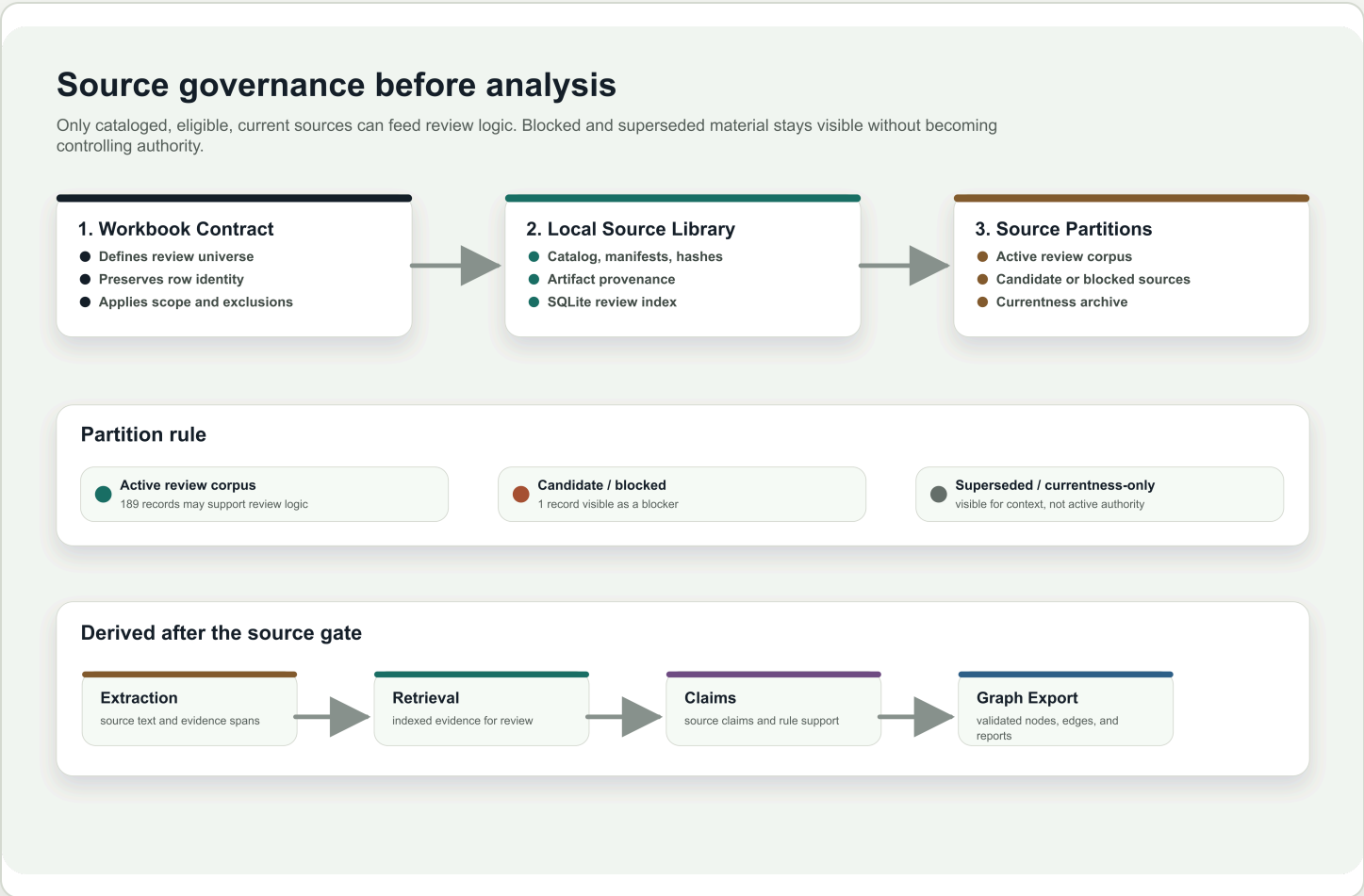
Architecture components at a glance

A source-governed review engine with traceable evidence, explicit authority logic, and gated outputs.



Controlled source library to graph-ready review layer

Every review starts with a controlled source contract. Workbook rows, exclusions, URL repairs, source partitions, catalog records, and artifact hashes define what can support review logic. Derived layers are rebuilt from those surfaces, not loose files.

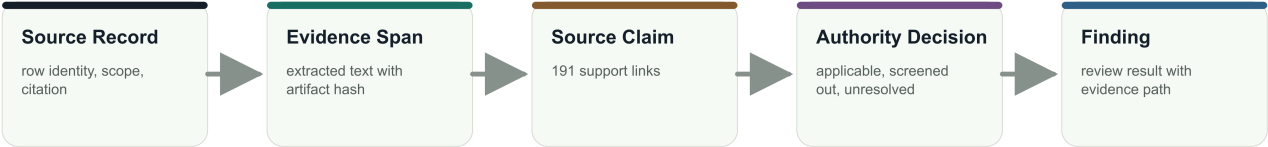


Every finding stays connected to evidence

Findings are produced from inspectable evidence paths. The system links source records, artifacts, evidence spans, claims, authority decisions, and review findings before it renders reports or graph views.

Finding trace

The review product is only the final view. The underlying path stays inspectable from source record to finding.



Reverse compliance uses the same path in the opposite direction.
Unsupported claims, stale authority language, missing evidence, and unresolved applicability are flagged before readiness.

Outputs generated from the trace



Traceability model: the finding is the end of a chain, not a free-standing conclusion. The same chain supports reports, QA replay, and graph exploration.

Audit surface

Each product can be traced back to source record, artifact hash, evidence span, and decision path. Reviewers can inspect the basis without relying on narrative alone.

Reverse compliance

The evidence model flags unsupported claims, older-regulation dependencies, missing source support, unresolved applicability, and forest-plan gaps before readiness is claimed.

Promotion gates make readiness explicit

A polished report is not the readiness signal. The system checks source capture, currentness, evidence, graph validity, review outputs, final QA, and promotion before it marks a review ready.

Readiness gate stack

Readiness is an artifact state. The system must prove source, evidence, graph, review, QA, and promotion gates before outputs are treated as ready.

Current readiness gates

	Source capture catalog and artifact links complete	PASS
	Currentness active, blocked, and superseded roles checked	PASS
	Evidence build extraction, retrieval, claims, and graph inputs rebuilt	PASS
	Graph validation schema, provenance, endpoints, and summaries valid	PASS
	Review outputs matrix, decision support, final QA, and PDF artifacts valid	PASS
	Promotion current gate suite ready	PASS

Expansion blockers stay typed

Forest profile not ready 62 profile coverage not complete
Missing source 33 required source absent or unresolved
Superseded source 3 visible for currentness only
Chapter delta 2 handbook chapter split required

1/10 tracked forest-plan profiles are graph-ready.

Ready means replayable

The underlying source, evidence, graph, review, QA, and promotion artifacts can be regenerated and checked against the same gates.

Blocked means specific

Missing sources, profile gaps, superseded authority, and chapter-level deltas stay typed and visible instead of becoming a generic failure.

Boundary statement: the system produces auditable review support and readiness evidence. It does not replace professional judgment, agency review, or legal sufficiency review.